

How to Find a δ for a given f , L , c , and $\epsilon > 0$

Steps:

1st step: Solve the inequality $|f(x) - L| < \epsilon$ to find an open interval (a, b) containing c on which the inequality holds for all $x \neq c$.

2nd Step: Find a value of $\delta > 0$ that places the open interval $(c - \delta, c + \delta)$ centered at c inside the interval (a, b) .

The inequality $|f(x) - L| < \epsilon$ will hold for all $x \neq c$ in this δ -interval.

Example: Prove that $\lim_{x \rightarrow 2} f(x) = 4$ if

$$f(x) = \begin{cases} x^2, & x \neq 2 \\ 1, & x = 2 \end{cases}$$

Solⁿ: We want to show that given $\epsilon > 0 \exists \delta > 0$ s.t.

$$|f(x) - 4| < \epsilon \text{ whenever } 0 < |x - 2| < \delta.$$

Solve the inequality $|f(x) - 4| < \epsilon$ to find an open interval containing $x = 2$ on which the inequality holds for all $x \neq 2$.

For $x \neq c = 2$, we have $f(x) = x^2$, we have

$$|x^2 - 4| < \epsilon$$

$$- \epsilon < x^2 - 4 < \epsilon$$

$$4 - \epsilon < x^2 < 4 + \epsilon$$

$$\sqrt{4-\epsilon} < |x| < \sqrt{4+\epsilon}$$

$$\sqrt{4-\epsilon} < x < \sqrt{4+\epsilon}$$

The inequality $|f(x)-4| < \epsilon$ holds for all $x \neq 2$ in the interval $(\sqrt{4-\epsilon}, \sqrt{4+\epsilon})$, assume $\epsilon < 4$.

2nd step

Now find a value of $\delta > 0$ that places the centered interval $(2-\delta, 2+\delta)$ inside the interval $(\sqrt{4-\epsilon}, \sqrt{4+\epsilon})$

Take δ to be the distance from $x=2$ to the nearer endpoint of $(\sqrt{4-\epsilon}, \sqrt{4+\epsilon})$.

i.e., ~~Take $\delta = \min \{ \sqrt{4-\epsilon}, \sqrt{4+\epsilon} \}$~~

Take $\delta = \min \{ 2 - \sqrt{4-\epsilon}, \sqrt{4+\epsilon} - 2 \}$

$\therefore \forall x$ $|f(x)-4| < \epsilon$ whenever $0 < |x-2| < \delta$, say $\epsilon < 4$.

if $\epsilon \geq 4$ then take $\delta = \min \{ 2, \sqrt{4+\epsilon} - 2 \}$.

4. Problem: $\lim_{x \rightarrow 0} \sqrt{4-x} = 2$ using ϵ - δ definition.

Solⁿ $\forall \epsilon > 0 \exists \delta > 0$ s.t. $0 < |x-0| < \delta$ whenever $|\sqrt{4-x} - 2| < \epsilon$.

$$\text{Now, } |\sqrt{4-x} - 2| = \left| \frac{(\sqrt{4-x} - 2)(\sqrt{4-x} + 2)}{(\sqrt{4-x} + 2)} \right|$$

$$= \left| \frac{4-x-4}{\sqrt{4-x}+2} \right|$$

$$= \frac{|x|}{|\sqrt{4-x}+2|}$$

$$\left\{ \begin{array}{l} x \leq 4 \\ 4-x \geq 0 \\ \sqrt{4-x} \geq 0 \\ \sqrt{4-x} + 2 \geq 2 \end{array} \right.$$

Since $\sqrt{4-x} \geq 0$ for $x \leq 4$, then $\sqrt{4-x} + 2 \geq 2$

$$\Rightarrow \frac{1}{\sqrt{4-x}+2} \leq \frac{1}{2}$$

$$\text{Therefore } \frac{|x|}{|\sqrt{4-x}+2|} \leq \frac{|x|}{2}$$

Now for any $\epsilon > 0$, we define $\delta = \min\{2\epsilon, 4\}$

Since $|x| < 4$ already defined.

Now, for $|x| < 2\epsilon = \delta$

$$|\sqrt{4-x} - 2| = \frac{|x|}{\sqrt{4-x}+2} \leq \frac{|x|}{2} < \frac{2\epsilon}{2} = \epsilon$$

which implies that $|\sqrt{4-x} - 2| < \epsilon$. $\square$

Why $\delta = \min\{2\epsilon, 4\}$?

Since $\sqrt{4-x}$ is only defined for $x \leq 4$. By setting $\delta \leq 4$, we guarantee that for all x satisfying $|x-0| < \delta$.

For the inequality $\frac{|x|}{\sqrt{4-x}+2} \leq \frac{|x|}{2}$, $\frac{|x|}{2} < \epsilon \Rightarrow |x| < 2\epsilon$. Then $\delta \leq 2\epsilon$.

Using ϵ - δ definition:

$$1. \lim_{x \rightarrow 4} (9-x) = 5$$

Solⁿ $\forall \epsilon > 0 \exists \delta > 0$ s.t. $|9-x-5| < \epsilon$ whenever $0 < |x-4| < \delta$.

$$\text{Now } |9-x-5| < \epsilon$$

$$|4-x| < \epsilon$$

$$|-(x-4)| < \epsilon$$

$$|x-4| < \epsilon$$

Thus, we take $\delta = \epsilon$

Verification:
If

$0 < |x-4| < \delta = \epsilon$, then $|(9-x)-5| = |4-x| = |x-4| < \delta = \epsilon$
i.e. $|9-x-5| < \epsilon$.
*.

$$2. \lim_{x \rightarrow 1} f(x) = 1 \text{ if } f(x) = \begin{cases} x^{\sim}, & x \neq 1 \\ 2, & x = 1 \end{cases}$$

Solⁿ We want to show that $\forall \epsilon > 0 \exists \delta > 0$ s.t.

$$|f(x)-1| < \epsilon \text{ whenever } 0 < |x-1| < \delta.$$

Since $x \neq 1$ for $f(x) = x^{\sim}$, we have

$$|x^{\sim}-1| < \epsilon$$

$$-\epsilon < x^{\sim}-1 < \epsilon$$

$$1-\epsilon < x^{\sim} < 1+\epsilon$$

$$\sqrt{1-\epsilon} < x < \sqrt{1+\epsilon}$$

The inequality $|f(x)-1|<\epsilon$ holds for all $x \neq 1$ in the interval $(\sqrt{1-\epsilon}, \sqrt{1+\epsilon})$ where assume $\epsilon < 1$.

Now find a $\delta > 0$ s.t. $(1-\delta, 1+\delta)$ inside the interval $(\sqrt{1-\epsilon}, \sqrt{1+\epsilon})$.

Take δ to be the distance from $x=1$ to the nearer endpoint of $(\sqrt{1-\epsilon}, \sqrt{1+\epsilon})$ i.e.

$$\text{Take } \delta = \min \{1-\sqrt{1-\epsilon}, \sqrt{1+\epsilon}-1\}$$

So $\forall x$ $|f(x)-1|<\epsilon$ whenever $0 < |x-1| < \delta$, $\epsilon < 1$

If $\epsilon \geq 1$ then take $\delta = \min \{1, \sqrt{1+\epsilon}-1\}$.

✱

How to prove that $\lim_{x \rightarrow c} f(x) \neq L$. [Showing L is not a limit]

Sol. We want to prove that $\lim_{x \rightarrow c} f(x) \neq L$ by providing an $\epsilon > 0$ such that no possible $\delta > 0$ satisfies the condition

$$|f(x) - L| < \epsilon \text{ whenever } 0 < |x - c| < \delta.$$

For each $\delta > 0$ $\exists x$ such that

$$0 < |x - c| < \delta \text{ and } |f(x) - L| \geq \epsilon. \quad [\text{using negation}]$$

Problem: Prove that $\lim_{x \rightarrow 3} (x^2 + 4) \neq 5$.

Sol. Using negation:

$\exists \epsilon > 0$ such that $\forall \delta > 0$, there is an x with $0 < |x - 3| < \delta$ but $|(x^2 + 4) - 5| \geq \epsilon$.

Choose $\epsilon = 1$

For any $\delta > 0$ select $x = 3 + \frac{\delta}{2}$

$$\text{so, } 0 < |x - 3| = \left| 3 + \frac{\delta}{2} - 3 \right| = \frac{\delta}{2} < \delta.$$

$$\begin{aligned} \text{Now, } |(x^2 + 4) - 5| &= \left| \left(3 + \frac{\delta}{2} \right)^2 + 4 - 5 \right| = \left| 9 + 3\delta + \frac{\delta^2}{4} - 1 \right| \\ &= \left| 8 + 3\delta + \frac{\delta^2}{4} \right| \end{aligned}$$

$$\text{since } \left| 8 + 3\delta + \frac{\delta^2}{4} \right| > 8 > 1 = \epsilon.$$

$$\text{Thus, } |(x^2 + 4) - 5| \geq \epsilon.$$

□

Some Extra problems:

Using ϵ - δ definition:

1. $\lim_{x \rightarrow 4} (9-x) = 5$

2. $\lim_{x \rightarrow 1} f(x) = 1$ if $f(x) = \begin{cases} x^2, & x \neq 1 \\ 2, & x = 1 \end{cases}$

3. $\lim_{x \rightarrow 0} \sqrt{4-x} = 2$

4. $f(x) = 3-2x$, $c = 3$, $\epsilon = 0.02$

5. ~~$\lim_{x \rightarrow 1} f(x) = 2$ if $f(x) = \begin{cases} 4-2x, & x < 1 \\ 6x-4, & x > 1 \end{cases}$~~

5. $\lim_{x \rightarrow -2} f(x) = 4$ if $f(x) = \begin{cases} x^2, & x \neq -2 \\ 1, & x = -2 \end{cases}$

6. let $h(x) = \begin{cases} (x^2+1)^2, & x \neq 2 \\ 2, & x = 2 \end{cases}$

Show that $\lim_{x \rightarrow 2} h(x) \neq 4$